

Closing the Pre-Infusion Gap: A Quality Improvement Initiative to Enhance Laboratory Readiness for Outpatient IV Iron Referrals

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Laboratory completion increased from 69.1% to 89.8%

Laboratory treatment-parameter attainment remained high: 90.8% at baseline and 94.9% after PDSA 2.

MOTIVATION

At Renown in Northern Nevada, some patients arrived for IV iron infusions without valid iron studies, requiring same-day testing and coordination among nursing staff, pharmacy, and referring clinicians. Lab orders embedded in treatment plans did not reliably ensure readiness before arrival, highlighting the need for earlier verification.

★ **Nearly 1 in 3 lacked valid iron studies before infusion**
Baseline: 64/207 | Jan–Apr 2026

AIM

Increase required iron study completion to ≥85% within six months among internal IV iron referrals, supported by verification before scheduling.

ESTIMATED CHAIR-TIME BURDEN

Baseline period: Jan–Apr 2026
Assuming 75 additional chair-minutes per visit lacking valid iron studies

- 20 additional chair-hours/month
- 240 chair-hours/year (annualized)*

**Based on an infusion pharmacy staff estimate of 75 extra minutes per affected visit. Annual projection assumes unchanged baseline volume and frequency of missing or invalid studies; delays were not directly measured.*

METHODS

Single-center QI evaluation of 416 internal outpatient IV iron referral records, January 1–September 14, 2026. All records were included regardless of treatment status.

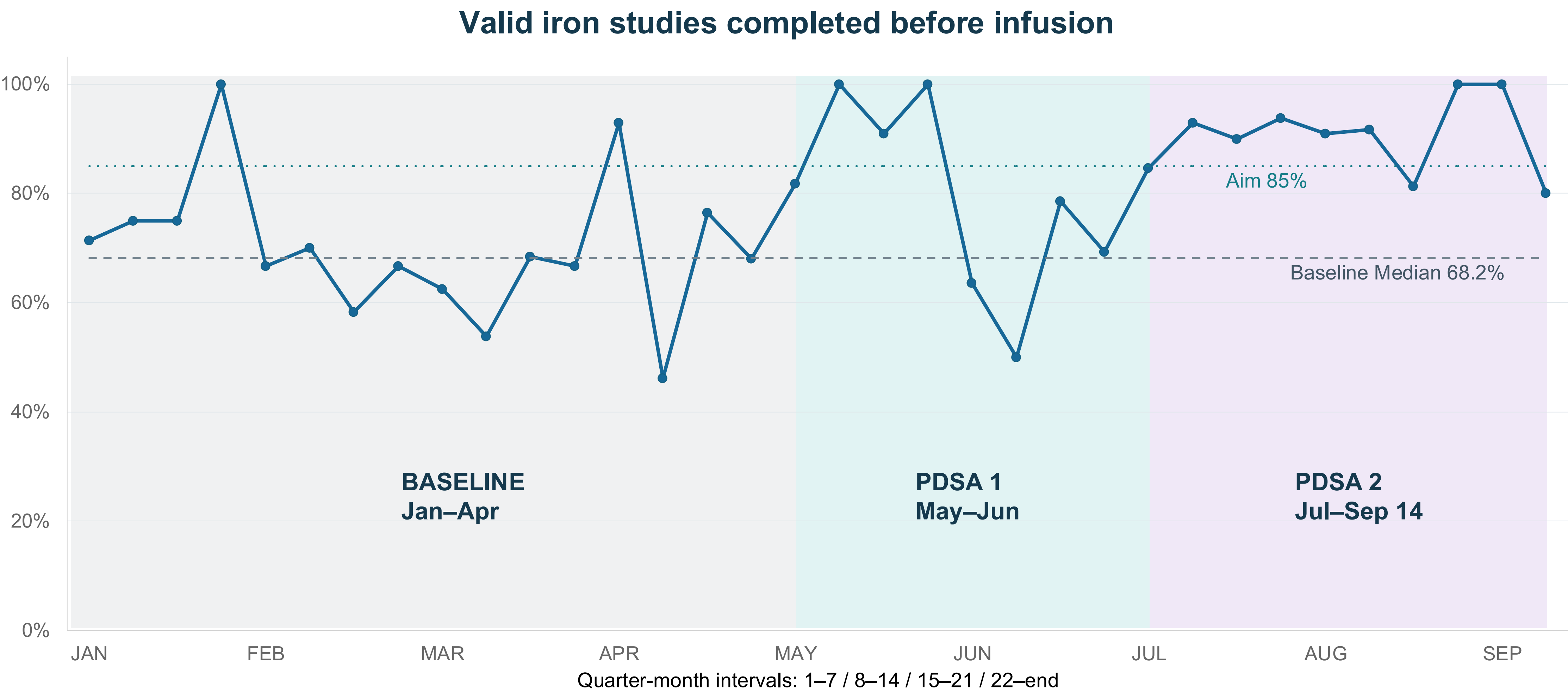
Process measure: Percentage with complete, valid iron studies before infusion.

Outcome measure: Percentage meeting laboratory treatment parameters.

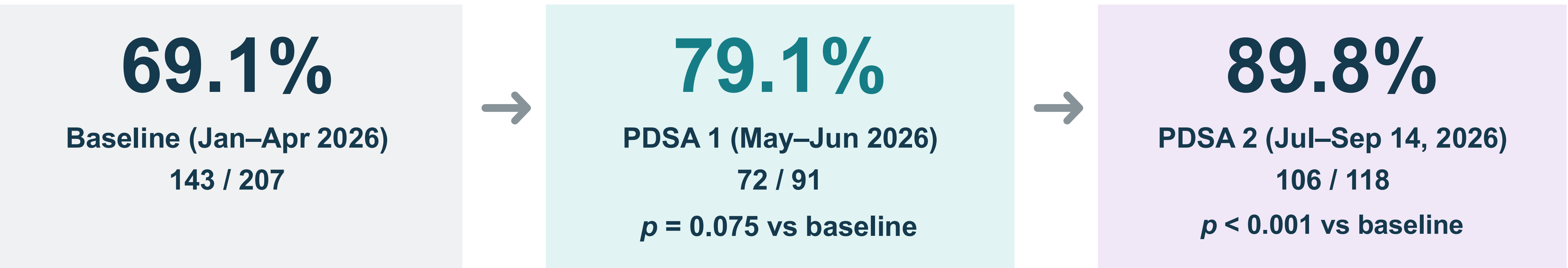
Analysis:

- Quarter-month run chart displayed process performance over time using a fixed baseline median.
- Pearson’s chi-square test compared pooled laboratory readiness rates across the three study periods: baseline, PDSA 1, and PDSA 2, assuming independent records. Outcome rates were summarized descriptively.

PROCESS MEASURE: LABORATORY READINESS OVER TIME

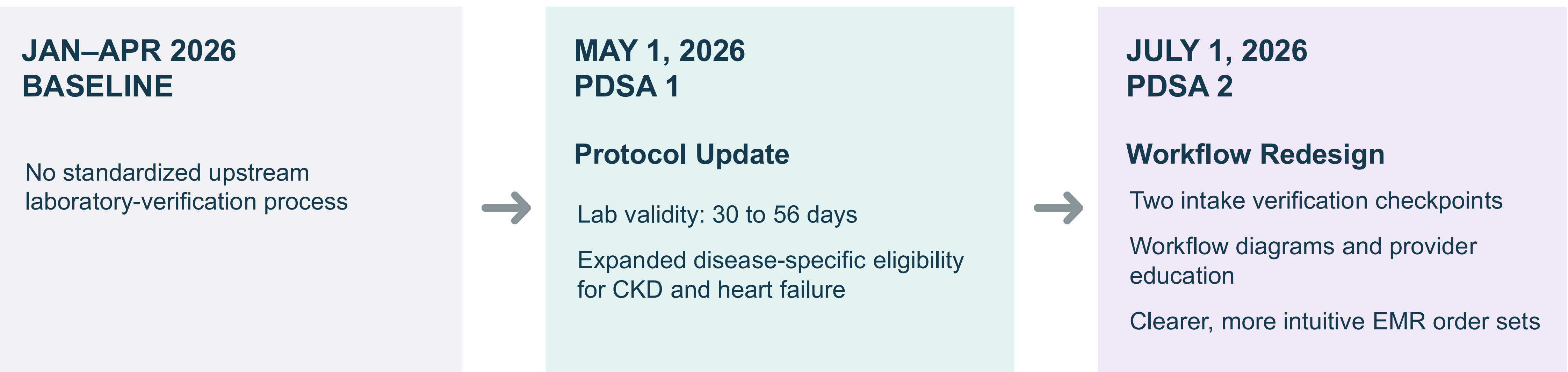


LABORATORY READINESS BY PERIOD



Overall $p < 0.001$ across three periods.
Pairwise p-values compare each intervention period with baseline using Pearson’s chi-square tests with Holm adjustment.

WHAT WE CHANGED



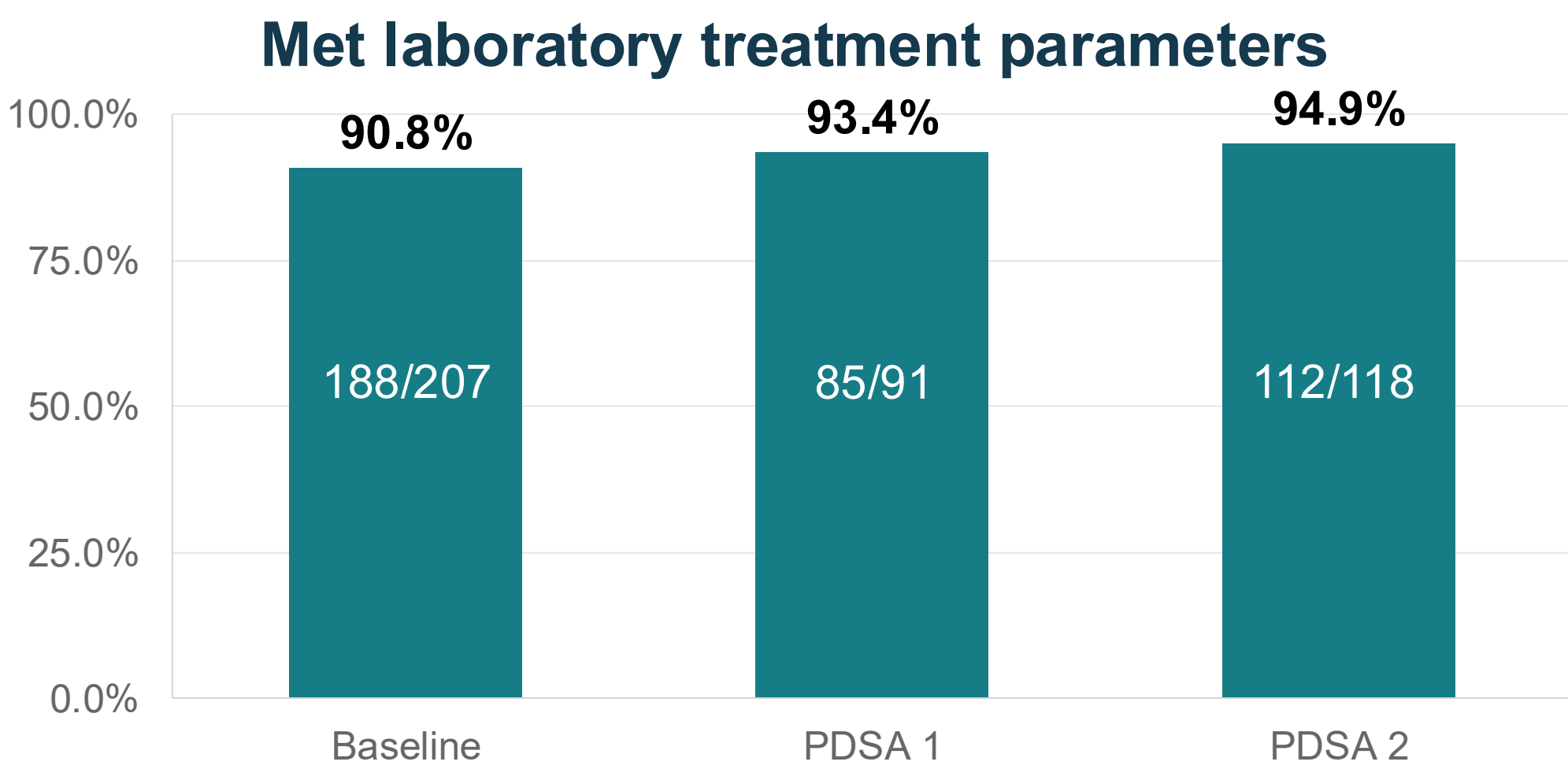
ACKNOWLEDGMENTS

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REFERENCES

Renown Health. *Pricing Transparency*. Accessed September 30, 2026. <https://www.renown.org/Patients-and-Visitors/Billing/Pricing-Transparency>

OUTCOME MEASURE



Attainment remained high across all periods.

PATIENT IMPACT (ESTIMATED)

≈161 patient-hours of extra chair time potentially avoided annually. *

Potentially less waiting, fewer treatment disruptions, and less uncertainty for patients and caregivers, including those traveling from rural Northern Nevada.

**Uses the same chair-time model as the capacity estimates below. Patient benefits were not directly measured. Baseline compared with PDSA 2*

MODELED ANNUAL CAPACITY BURDEN

Modeled annual burden	Baseline	PDSA 1	PDSA 2
Extra chair-hours/year*	240	162	79
Illustrative annual value†	\$279,000	\$188,000	\$92,000

≈161 chair-hours/year potentially available for other infusion therapies.
* Assumes 621 visits/year, 75 extra minutes per affected visit (staff estimate), and sustained rates; delays unmeasured.
† Uses the CPT 96365 cash price as an unvalidated \$1,161.30/hour proxy; not actual costs or savings.
Capacity gains depend on staffing, scheduling, and demand.

LIMITATIONS

Single-center, uncontrolled evaluation with limited follow-up. Changes in laboratory validity and eligibility criteria may affect comparisons across periods. Individual intervention effects cannot be isolated. Chair-time and financial estimates are modeled, not directly measured.

CONCLUSION

Protocol and workflow changes were associated with improved laboratory readiness, exceeding the 85% target overall after PDSA 2. Laboratory treatment-parameter attainment remained high. Intermittent below-target intervals highlight the need for continued monitoring and refinement.

NEXT STEPS

Monitor sustainability and adapt the workflow for an external-referral pilot.

